SECTION 2: PROJECT PLAN

Banking Transaction System – Requirements & Design Specification

2. Project Plan

2.1 Work Breakdown Structure (WBS)

The project scope is deconstructed into standard execution phases aligned with the system modules:

- **Phase 1: Project Setup & Architecture**
  - Initialize Git repository, Spring Boot 3.5.x skeleton, Maven configuration, and dev seed data.
  - Validate SQLite connection pooling, WAL mode, and single-writer immediate transaction architecture spike.
- **Phase 2: Authentication & User Management (AUTH, CUS)**
  - Implement customer registration, login endpoints, session handling, and Basic API authentication.
  - Configure BCrypt password hashing, security filter chain, and role-based access control (CUSTOMER, ADMIN).
  - Develop customer profile views and account overview dashboard.
- **Phase 3: Core Banking Operations & Ledger (ACC, TXN)**
  - Build domain model classes for Savings, Checking, and Fixed Deposit account types.
  - Implement deposit, withdrawal, and transfer service methods with atomic balance updates.
  - Enforce transaction limits, account withdrawal rules, dynamic transfer fee calculations, and append-only ledger entries.
- **Phase 4: Reporting, Administration, & Scheduling (STM, ADM, JOB, AUD)**
  - Create account statement views with pagination, date filtering, and CSV export functionality.
  - Build admin dashboard metrics, customer search, account freeze/unfreeze endpoints, and audit logging.
  - Implement scheduled jobs for monthly savings interest accrual and fixed deposit maturity processing.
- **Phase 5: Automated & Security Testing**
  - Develop unit tests for domain rules and MockMvc integration tests for REST controllers.
  - Build Postman API test collection and Selenium 4 WebDriver UI test scripts using stable element IDs.
  - Execute security testing targeting OWASP Top 10 risks (IDOR, privilege escalation, XSS, CSRF).
- **Phase 6: Quality Assurance & Final Deliverables**
  - Conduct static code analysis with SonarLint / SpotBugs and perform OWASP dependency vulnerability checks.
  - Resolve bugs logged in Jira, generate JaCoCo code coverage reports, and finalize technical documentation.

2.2 Timeline & Milestones

The 8-week schedule defines key delivery dates, dependencies, and project milestones:

- **Schedule Overview:**
  - **Week 1:** Requirements signoff, repository setup, database schema, SQLite spike validation, and test plan start.

- **Week 2:** Domain classes, unit tests, user registration, authentication, and role management.
- **Week 3:** Accounts, deposits, withdrawals, transfers, REST API endpoints, UI forms, and error handling.
- **Week 4:** Account statements, CSV export, admin dashboard, audit log, scheduled jobs, and UI element ID freeze.
- **Week 5:** Selenium UI automation, Postman API collection, and security test cases.
- **Week 6:** Static testing (SonarLint/SpotBugs), bug fixing, and full regression testing.
- **Week 7:** OpenAPI/Swagger documentation, Jira test-activity reports, JaCoCo coverage analysis, and demo rehearsal.
- **Week 8:** Buffer week and final project presentation/defense.

- **Key Project Milestones:**
  - **Milestone 1 (Week 1):** Project Foundations & Architecture Signoff (Database schema, seed data, SQLite spike verified).
  - **Milestone 2 (Week 3):** Core Banking MVP (Complete transaction engine with fee calculation and ledger logging).
  - **Milestone 3 (Week 4):** Feature Complete & UI Freeze (Admin tools, statements, frozen DOM IDs for automation).
  - **Milestone 4 (Week 7):** Code Quality & Test Signoff (Automated tests passing, documentation complete, zero critical bugs open).
  - **Milestone 5 (Week 8):** Final System Delivery & Presentation.

2.3 Resource Plan

The required technology stack, developer tools, hardware, and runtime environments are detailed below:

Resource Category	Tools & Environment Details
Backend Framework	Java 17 or higher, Spring Boot 3.5.x (Web, Data JPA, Security, Validation), Maven
Database & Storage	SQLite (embedded single-file database), sqlite-jdbc, Hibernate SQLite dialect, DB Browser for SQLite
Frontend & UI	Thymeleaf view templates, plain CSS (or Bootstrap framework)
Testing Tools	JUnit 5, Mockito, Spring Boot Test / MockMvc, Selenium 4 (Page Object Model), Postman, JaCoCo
Quality & Process	Git + GitHub (pull requests), Jira (board, bug tracking), SonarLint / SpotBugs, OWASP Dependency-Check
Hardware / Host	Standard developer workstations running JDK 17+ and local embedded database environment